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PROJECT: PARKINSON'S DISEASE EARLY DIAGNOSIS USING MACHINE LEARNING AND LLM

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2. BACKGROUND AND MOTIVATION

Parkinson's Disease (PD) is a progressive neurodegenerative disorder that affects movement and cognitive function, impacting over **10 million people** worldwide. Despite its prevalence, there is currently no cure, and early detection remains a significant barrier to effective disease management. Clinical diagnosis is often delayed until symptoms become pronounced, by which time significant neurological damage may have already occurred.

Traditional diagnostic methods rely heavily on clinician-administered questionnaires, neurological examinations, and patient history. These assessments are not only time-consuming and resource-intensive but also suffer from subjectivity and inconsistent interpretations. In countries like India, where healthcare access is uneven and linguistic diversity is immense, such conventional methods become difficult to scale and deploy effectively.

Meanwhile, recent advances in Machine Learning (ML) and Large Language Models (LLMs) offer promising pathways for automating complex cognitive tasks, including health-related dialogue and decision-making. By simulating natural human conversation, LLMs can administer and interpret self-screening questionnaires, potentially serving as virtual clinical assistants.

The PPMI Online dataset, with its large-scale, self-reported responses and real-world data structure, provides a unique opportunity to develop and train such models. Leveraging this dataset can lead to the creation of a multilingual, accessible, and intelligent early screening tool for Parkinson's Disease, empowering users to engage in proactive health assessments from the comfort of their homes.

This project is driven by the vision of democratizing access to early diagnostic tools and supporting public health in underserved regions through technology-enabled solutions

3. PROBLEM STATEMENT, OBJECTIVES AND SCOPE

3.1 PROBLEM STATEMENT

Parkinson's Disease (PD) is a chronic, progressive, and currently incurable neurodegenerative disorder that affects approximately 2 out of every 1000 individuals globally. Early diagnosis of Parkinson's is critical to managing symptoms and improving quality of life, yet it remains a significant challenge.

Conventional clinical diagnosis relies heavily on comprehensive, time-intensive assessments and subjective interpretations by medical professionals. These methods are often impractical for large-scale implementation, especially in resource-constrained settings such as India.

Furthermore, the linguistic and cultural diversity in India adds complexity to standardized clinical assessments, making scalable early screening even more difficult.

Given these challenges, there is a pressing need for an accessible, scalable, and language-agnostic screening tool. The objective of this project is to develop a Large Language Model (LLM)-based self-screening tool capable of engaging individuals in natural language conversations. The tool will ask clinically relevant questions, interpret responses, and assist in the early detection of Parkinson's Disease, thus enabling timely medical intervention.

3.2 OBJECTIVE

❖ Develop an LLM-Based Self-Screening Tool

Design a Large Language Model (LLM) system capable of engaging users in natural language conversations to screen for early symptoms of Parkinson's Disease.

❖ Leverage PPMI Online Data

Utilize the PPMI Online dataset, which includes large-scale, self-reported questionnaire data, to train and validate machine learning and deep learning models for symptom detection.

❖ Incorporate Missingness Tolerance

Design the model to handle incomplete or skipped responses gracefully, reflecting real-world user behaviour and improving model robustness.

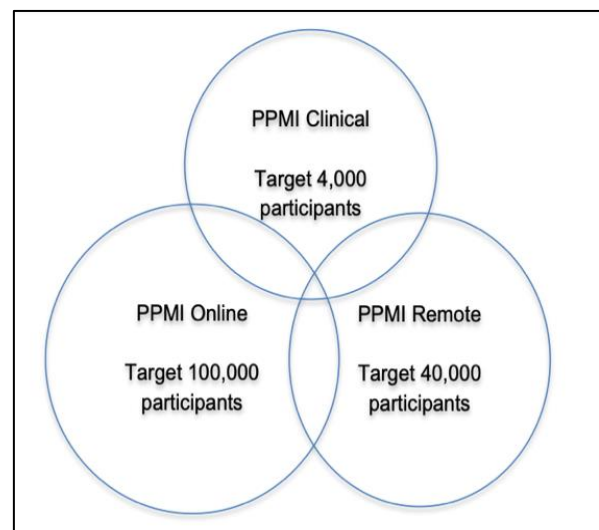
3.3 SCOPE

- ❖ Focused on **early-stage Parkinson's detection** using questionnaire-style inputs.
- ❖ Covers **text-based interaction only**; voice, imaging, or sensor-based assessments are outside this project's scope.
- ❖ Trained and evaluated on **PPMI Online dataset only** for this phase.
- ❖ **Model interpretability** and explainability will be considered for user trust.
- ❖ The final deliverable will be a **prototype self-screening interface** (web-based or chatbot) supported by a trained LLM backend.

3.4 WHY PPMI ONLINE

To build an effective and scalable tool for early diagnosis of Parkinson's Disease, it is crucial to train and validate models on large, diverse, and representative datasets. The **Parkinson's Progression Markers Initiative (PPMI)** provides such data through two primary components:

- **PPMI Clinical:** Focused on in-depth, in-person clinical assessments with around 4,000 participants.
- **PPMI Online:** A large-scale, self-reported dataset designed for broader and more scalable research applications.



Key Advantages of PPMI Online:

- **Large Sample Size:** Aims to reach approximately **100,000 participants**, offering significantly greater statistical power compared to traditional clinical datasets.
- **Real-World Data:** Captures **self-reported responses** that reflect natural variability and missingness—mimicking actual deployment conditions of digital tools.
- **Multi-Timepoint Coverage:** Includes data from both initial assessments and **follow-up visits**, enabling longitudinal analysis and model refinement over time.

- **Ideal for ML/LLM Training:** Its scale and diversity make it particularly well-suited for **training machine learning and deep learning models** aimed at **early detection** of Parkinson's through questionnaire-based or conversational inputs.

PPMI Online serves as a critical foundation for building a self-screening LLM application that reflects the real-world challenges and behaviours of a diverse population, especially in contexts like India.

4. METHODOLOGY

4.1 DATA PREPROCESSING

Data preprocessing is a critical step in any machine learning pipeline, especially in healthcare analytics where data quality and relevance directly influence the outcomes. For our project, we worked with a rich clinical dataset composed of multiple CSV files containing extensive questionnaire responses from patients.

4.1.1 SELECTION CRITERIA

To ensure that our analysis focused on the earliest and most informative patient responses, we filtered the data to include only entries from the **first visit**, referred to as the **"Screening" event**. This strategic decision enabled us to capture patient responses *before any formal diagnosis* or treatment had potentially influenced their self-reporting behaviour or medical status. By doing so, our model could focus on recognizing subtle early signals indicative of Parkinson's Disease (PD).

4.1.2 INITIAL DATASET OVERVIEW

The initial dataset consisted of **890 columns**, each corresponding to a different questionnaire item, and several individual CSV files organized by questionnaire type and event. Merging these files required aligning on two primary keys: **Patient ID** and **Event ID**. We programmatically combined these into a unified tabular structure using Python-based data processing tools such as **pandas**.

4.1.3 DATA CLEANING AND FILTERING

Given the sparsity often associated with large-scale medical questionnaires, we applied the following filters:

- **Event Filter:** We retained only data associated with the "Screening" event.
- **Missing Data Threshold:** Columns with more than **50% missing values** were removed to preserve data integrity and reduce noise.
- **Irrelevant Features Removal:** Columns that did not contribute to the clinical assessment (such as timestamps or metadata fields) were discarded.

After the cleaning phase, the dataset was reduced to **143 informative columns**, significantly improving manageability while retaining valuable signals. In total, we were left with **38,256 patient-level records**, offering a comprehensive view of self-reported information and early-stage symptom data.

4.2 CREATION OF DATASET FOR LARGE LANGUAGE MODELS (LLMs)

To harness the capabilities of Large Language Models (LLMs), such as GPT, we restructured the tabular data into a prompt-based format that these models could effectively understand and process.

4.2.1 MAPPING QUESTIONS TO PROMPTS

Each of the 143 questionnaire columns was treated as a unique question, and a corresponding descriptive keyword was manually or semi-automatically assigned to each for clarity. Using few-shot prompting techniques with GPT, we generated full-form natural language question texts for each column. This transformation was essential to improve interpretability and ensure the final prompts resembled realistic patient-clinician dialogues.

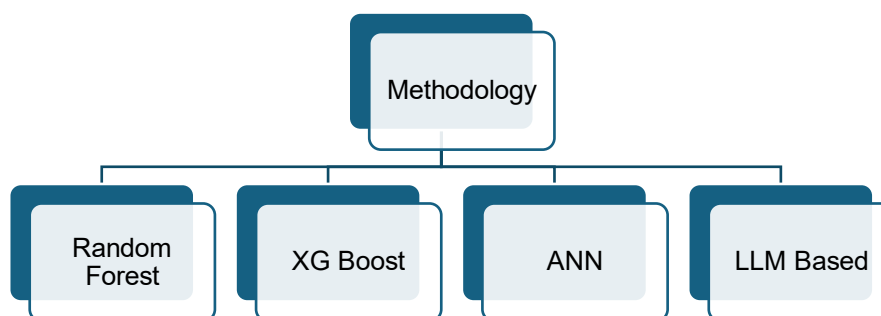
4.2.2 PATIENT-LEVEL DATA TRANSFORMATION

For each patient, the responses were converted into a structured dictionary format with the following two components:

- **prompt:** A concatenated string consisting of all the questions that the patient had answered, followed by their respective responses. This format simulates a conversation or clinical intake session.
- **label:** The diagnosis status of the patient, typically binary, indicating whether the patient was diagnosed with Parkinson's Disease (PD) or not.

4.2.3 BENEFITS OF THIS REPRESENTATION

- Allows the model to learn contextual patterns from conversational text rather than sparse vectors.
- Preserves the semantic meaning of the questions and responses.
- Offers flexibility in expanding to multilingual or cross-domain adaptations.
- Enables integration with prompt-tuning and few-shot learning techniques for low-data scenarios.



4.3 XGBOOST

Extreme Gradient Boosting (XGBoost) is a high-performance, scalable machine learning algorithm known for its superior accuracy in structured/tabular data tasks. In this project, XGBoost was crucial in the following ways:

4.3.1 Feature Importance Extraction

- XGBoost was used to compute **normalized feature importance scores** for every questionnaire item in the dataset.
- These scores helped identify the **top 30 most predictive features**, capturing **91.5% of the cumulative importance**.
- The selection allowed the project to retain maximum predictive power while significantly **reducing dimensionality**, leading to faster training and better generalization.

4.3.2 Scree Plot Visualization

- The output feature importances were visualized using a scree plot, which revealed the point of diminishing returns beyond the top 30 features.
- This analysis informed both the neural network architecture and LLM input design.

4.3.3 Benchmark Model

- XGBoost was trained and evaluated on the reduced dataset (top 30 features) using an **80:20 train-test split**.
- Its performance served as a **baseline** to compare the diagnostic accuracy of the LLM-based tool and Artificial Neural Network (ANN).
- XGBoost's ability to handle missing values internally and model complex interactions made it especially effective on real-world, imputed health data.

4.4 RANDOM FOREST

Random Forest is a robust ensemble learning method based on decision trees and bootstrap aggregation (bagging). It played a complementary and validating role in the project:

4.4.1 Data Imputation with MissForest

- The **MissForest** algorithm, based on Random Forest, was used for **imputing missing values** in the PPMI Online dataset.
- It provided high-quality estimates by averaging across multiple decision trees, enabling the dataset to be effectively used for modeling despite natural missingness.
- An **imputation accuracy of 39%** was achieved on 5% randomly masked data—demonstrating its suitability for questionnaire-style datasets.

4.4.2 Feature Importance Derivation

- Random Forest was also used to compute feature importance scores as a secondary method, validating the feature selection produced by XGBoost.
- This cross-model agreement improved confidence in the chosen feature subset.

4.4.3 Model Benchmarking

- A **Random Forest classifier** was trained on the selected 30 features and evaluated using an **80:20 train-test split**.
- It served as an additional **performance benchmark** for comparing against XGBoost, ANN, and the final LLM-based approach.

- Its interpretability and robustness make it a reliable choice for healthcare datasets with inherent variability.

4.5 ANN

Artificial Neural Networks (ANNs) are flexible deep learning models capable of capturing complex, non-linear relationships in data. In this project, the ANN served as a modern machine learning baseline to compare against traditional models and the LLM-based system.

4.5.1 Model Architecture

- The ANN model named Parkinson ANN is a deep feedforward neural network consisting of the following layers:
 - ❖ **Input Layer:** 30-dimensional input (corresponding to the number of features).
 - ❖ **Hidden Layers:**
 - Layer 1: Linear (30 → 128) + ReLU + Dropout (0.4)
 - Layer 2: Linear (128 → 64) + ReLU + Dropout (0.4)
 - Layer 3: Linear (64 → 32) + ReLU
 - ❖ **Output Layer:** Linear (32 → 1) + Sigmoid (for binary output).

The model was trained for **50 epochs** using the **AdamW optimizer** with a learning rate of **0.001** and **binary cross-entropy loss** (BCELoss) as the loss function. Training was accelerated using a GPU if available.

4.5.2 Model Architecture

Model Training and Evaluation: During training, both training and validation losses were recorded across epochs to monitor convergence. The training loss decreased steadily from **0.2469 in epoch 2** to approximately **0.1834 by epoch 40**, indicating successful optimization. Similarly, validation loss improved, stabilizing around **0.206–0.209** in later epochs.

After training, the model was evaluated on the test set using accuracy and a classification report. The sigmoid output of the model was thresholded at **0.5** to determine class labels.

4.5.3 Purpose in Methodology

- The ANN model was trained using an **80:20 train-test split** to evaluate its performance on the same reduced dataset used by XGBoost and Random Forest.
- It served as a **deep learning benchmark**, representing how well classical neural networks perform on structured questionnaire data.
- The results from ANN helped assess whether moving to more advanced LLM-based architectures could offer significant gains.

4.6 LLM

We chose Google's Flan-T5 Small, released in Oct 2022, for fine-tuning. It enhances generalization through instruction tuning. It was chosen for its:

1. Encoder-Decoder architecture (strong in classification and generalization)
2. Suitability for limited computational resources

3. Lightweight design (80 million parameters)

4. Superior performance on summarization, classification, and QA tasks.

The model's lightweight design and strong out-of-the-box performance on downstream tasks make it ideal for environments with limited computational resources/

To adapt Flan-T5 for binary classification, we appended a trainable linear layer on top of the mean-pooled final hidden states of the encoder. A dropout of 0.1 was applied before the classification projection to mitigate overfitting. During training, the base model weights were either fully frozen (in the quantized LoRA setup) or partially unfrozen, depending on the adaptation strategy.

In each strategy, the models were trained on both the complete dataset and the top 30 features dataset, each randomly split in a **75:15:15** train-validation-test split. We used early stopping based on validation loss with patient of 3 epochs:

End-to-End Fine Tuning:

We fine-tuned the complete model (including classifier parameters) end-to-end, updating all 80 million parameters. We used a batch size of 128 and a learning rate of 1×10^{-5} . Due to limited computation resources, the model was only trained the complete dataset.

LowRank Adaptation (LoRA) with quantization:

LoRA freezes the pretrained weights and injects trainable low-rank decomposition matrices into each attention weight matrix, so that

$$\Delta W = AB \quad \text{with } A \in \mathbb{R}^{d \times r}, B \in \mathbb{R}^{r \times k}$$

reducing the number of trainable parameters from $O(dk)$ to $O(r(d+k))$. We set the rank $r=16$ based on empirical practice that balances expressive capacity against memory footprint. Only the LoRA adapter parameters A and B are updated, yielding up to 95 % fewer trainable weights compared to full fine-tuning.

We implemented LoRA with rank 16 and backpropagation only through the adapter weights. We quantised the base model weights to 4 bits using the NormalFloat (NF4) format, reducing storage requirements by $\sim 4\times$ over 16-bit fine-tuning while preserving full precision task performance.

5. IMPLEMENTATION AND DEMONSTRATION

The proposed system was implemented in two main phases:

1. Machine Learning-Based Questionnaire Modelling, and
2. LLM-Based Conversational Screening Interface.

PHASE 1: MACHINE LEARNING-BASED QUESTIONNAIRE MODELLING

1. Data Preprocessing & Imputation

- **Dataset Used:** *PPMI Online*, containing self-reported questionnaire responses.
- **Missing Value Handling:**
 - Applied the **MissForest imputer** (Random Forest-based) to handle natural missingness in user responses.

- Achieved **39% imputation accuracy** on masked test data (5% randomly hidden entries).

2. Feature Engineering

- Used **XGBoost and Random Forest** to calculate normalized feature importances.
- Selected **Top 30 features** based on cumulative contribution (~91.5%) to prediction accuracy.

3. Model Training & Evaluation

- **XGBoost & Random Forest:**
 - Trained using 80:20 split.
 - Evaluated on both complete and reduced (top 30 feature) datasets.
 - Plotted accuracy vs. number of features used to select the optimal input subset.
- **ANN:**
 - Input: Top 30 features.
 - Architecture: $30 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$
 - Trained with **Adam optimizer, 50 epochs, batch size = 128, learning rate = 0.001**.
 - Compared on accuracy, precision, recall, and F1-score.

PHASE 2: LLM-BASED CONVERSATIONAL SCREENING TOOL

To enable scalable, language-flexible self-screening for Parkinson's Disease, we implemented a **Large Language Model (LLM)** using **Google's Flan-T5 Small**. Released in October 2022, Flan-T5 is an instruction-tuned encoder-decoder model well-suited for classification tasks and resource-constrained environments.

1. Model Selection: Flan-T5 Small

Flan-T5 Small was selected for the following reasons:

- **Encoder-Decoder architecture:** Effective for both understanding and generating text, especially in classification and QA tasks.
- **Lightweight (80M parameters):** Suitable for fine-tuning and deployment in environments with limited compute.
- **Superior generalisation:** Thanks to instruction tuning across a variety of NLP tasks.
- **Low resource adaptability:** Performs well with quantization and low-rank adaptation.

2. Model Adaptation for Classification

To adapt Flan-T5 for binary Parkinson's prediction:

- A **trainable linear classifier layer** was appended on top of the **mean-pooled encoder hidden states**.
- A **dropout layer (0.1)** was added before the projection layer to prevent overfitting.
- The final model outputs a binary prediction: *Parkinson's Risk vs No Risk*.

3. Training Strategies

Two different strategies were used to fine-tune the model:

A. End-to-End Fine-Tuning

- All **80M parameters** of Flan-T5 were updated during training.
- **Dataset:** Trained only on the **complete feature set** due to resource constraints.
- **Batch Size:** 128
- **Learning Rate:** 1×10^{-5}
- **Split:** 75% train / 15% validation / 15% test
- **Early Stopping:** Enabled (patience = 3 epochs)

This approach yielded a fully tuned model capable of learning deep representations, though it was computationally intensive.

B. LoRA with 4-Bit Quantization (QLoRA)

To reduce memory and computation demands:

- **Low-Rank Adaptation (LoRA)** was applied with **rank $r = 16$** , reducing trainable parameters by up to **95%**.
- Only LoRA matrices (A and B) in the attention layers were trained; **base model weights were frozen**.
- **4-bit quantization** using **NF4 (NormalFloat)** format was applied to the base weights.
 - Achieved $\sim 4\times$ reduction in memory over FP16
 - Preserved downstream task performance

Training Configuration:

- **Optimizer:** AdamW (weight decay = 0.01)
- **Learning Rate:** 1×10^{-4} (for adapter parameters)
- **Batch Size:** 32
- **Epochs:** Max 10 (early stopping after 3 epochs without improvement)

This setup allowed training on both **complete** and **top-30 feature** datasets, supporting rapid experimentation and deployment readiness.

4. Evaluation Metrics

For both strategies, model performance was assessed on the **held-out test set** using:

- **Accuracy**
- **Precision**
- **Recall**
- **F1 Score**

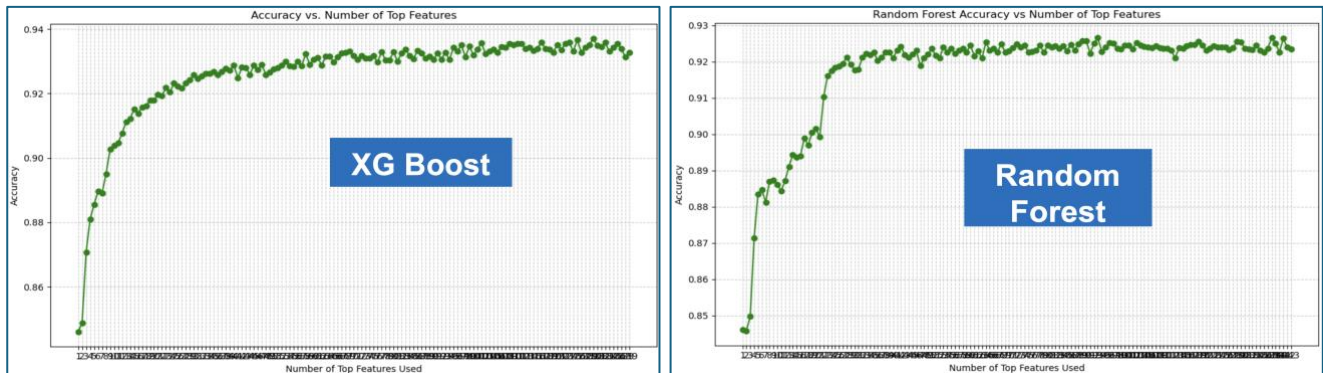
The LoRA+QLoRA pipeline offered a **computationally efficient** alternative with comparable performance to full fine-tuning, making it highly suitable for real-world low-resource deployment.

6. RESULTS

This section presents the evaluation outcomes of different machine learning and LLM-based models on the early-diagnosis questionnaire dataset. The models were assessed on their predictive performance using accuracy as the primary metric. Additionally, training and validation loss plots are presented to understand convergence behaviour across various configurations.

6.1 BENCHMARK MODELS (TRADITIONAL ML)

To establish a performance baseline, we trained three traditional machine learning models using both the **complete dataset (143 features)** and a **feature-engineered subset (top 30 features)**:



- **Random Forest (RF):**
 - Achieved **91.56%** accuracy on the full dataset.
 - Performance slightly decreased to **90.23%** with top 30 features, indicating that reduced dimensionality affected RF more than other models.
- **XG Boost:**
 - Delivered the **highest accuracy among benchmark models at 92.20%** using the full dataset.
 - Performance remained robust with **91.41%** on the reduced feature set.
- **Artificial Neural Network (ANN):**
 - Trained exclusively on the top 30 features and achieved **92.01%** accuracy.
 - Though it did not outperform XG Boost, its performance was closely competitive.

These models validate that even traditional machine learning approaches can perform strongly on structured questionnaire data, particularly when feature selection is optimized.

6.2 LARGE LANGUAGE MODEL-BASED APPROACHES

To explore more advanced modelling, we employed **FLAN-T5**, a powerful transformer-based large language model, using two fine-tuning strategies:

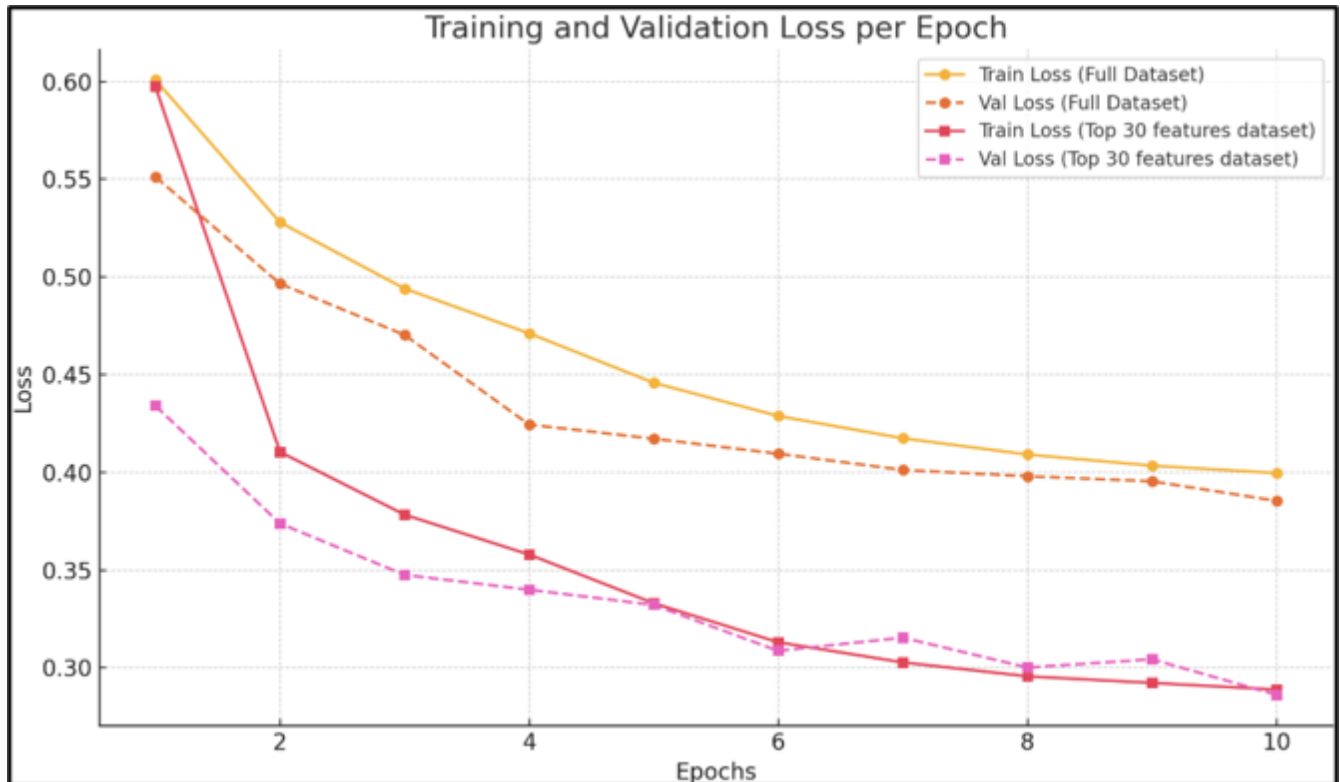
A. LORA-BASED FINE-TUNING

We utilized Low-Rank Adaptation (LoRA), a parameter-efficient technique to fine-tune the FLAN-T5-small model:

- **Accuracy with Full Dataset (143 Questions): 83.41%**
- **Accuracy with Top 30 Questions: 87.50%**

This demonstrates that even with reduced question complexity, the model could generalize better, possibly due to reduced noise and improved focus.

The accompanying **loss plots** (Shown Below) show steady convergence:



- The **training and validation loss** consistently decreased over 10 epochs.
- Models trained on the top 30 questions showed better generalization with a sharper loss reduction curve, indicating more efficient learning.

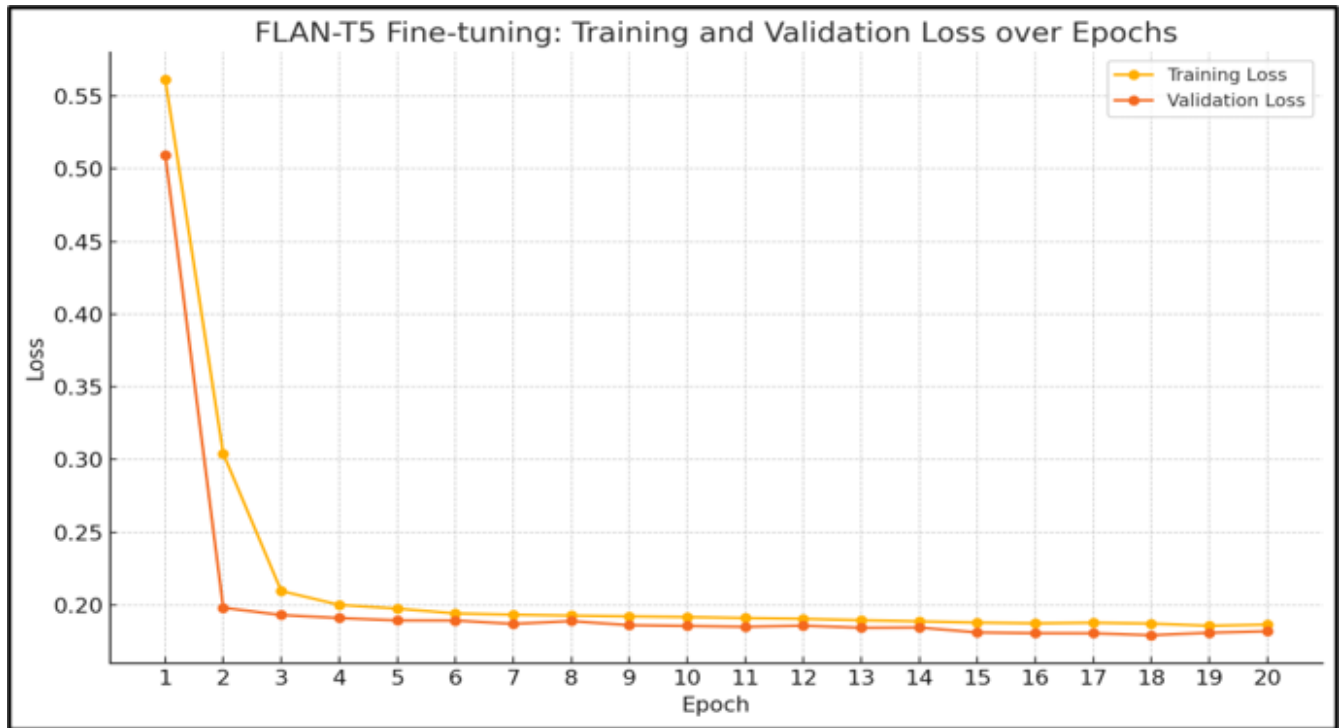
B. END-TO-END FINE-TUNING (FULL MODEL UPDATE)

We further fine-tuned the entire FLAN-T5 model end-to-end without parameter sharing or restriction:

- Achieved an **accuracy of 94.49%**, the highest across all models and configurations.
- Significantly outperformed traditional models and LoRA-based LLM tuning.

The Below plot reveals:

- Rapid convergence within the first few epochs, stabilizing around a **training loss of ~0.2** and validation loss showing similar behaviour.
- Minimal overfitting and strong generalization, making this method highly effective for prompt-based patient modelling.



7. CONCLUSION AND FUTURE SCOPE

7.1 RESULTS

The experimental evaluation compared both traditional machine learning models and advanced large language models (LLMs) in predicting disease diagnosis using structured questionnaire data. The primary focus was on understanding the impact of dataset size, feature selection, and fine-tuning approaches on model performance.

LLM with LoRA Adaptation:

The use of Low-Rank Adaptation (LoRA) with the FLAN-T5 small model showed considerable promise:

- **On the full dataset (143 questionnaires)**, the LLM with LoRA adaptation achieved an accuracy of **83.41%**. This performance was already competitive, considering the inherent complexity of natural language-based structured data and the minimal domain-specific customization at this stage.
- **On the reduced dataset with top 30 features**, the model's accuracy **improved to 87.50%**. This increase suggests that LoRA-enhanced LLMs are sensitive to signal-rich inputs, and that feature selection—removing noise and redundancy—can significantly enhance learning in such models. It confirms the model's ability to generalize better when focused on the most informative subset of the data.

Fine-Tuned LLMs Outperform Traditional Approaches:

A fully **end-to-end fine-tuned version of the FLAN-T5** model was also evaluated. This approach involved retraining the model using all available patient questionnaires without using LoRA.

- The **fine-tuned model achieved an accuracy of 94.49%**, which was the highest across all configurations. This outcome demonstrates that, when properly trained and adapted to the

domain-specific structure of the dataset, LLMs can outperform conventional machine learning algorithms even on moderately sized, structured datasets.

- These results validate the hypothesis that **fine-tuning enables LLMs to internalize domain-specific semantics and patterns**, surpassing manually engineered models that rely heavily on preprocessing and feature engineering.

Comparison with Traditional Models

Benchmark models including **XGBoost**, **Random Forest**, and **Artificial Neural Networks (ANN)** were also evaluated:

- **XGBoost** achieved the highest performance among traditional models with an accuracy of **92.20%** on the full dataset, followed closely by **ANN** (92.01%) and **Random Forest** (91.56%).
- When tested on the top 30 features, accuracies slightly decreased for these models but remained competitive. XGBoost retained **91.41%** accuracy while Random Forest dropped to **90.23%**, highlighting the robustness of gradient boosting.
- However, even the best-performing traditional model was outperformed by the **end-to-end fine-tuned LLM**, proving the advantage of deep representation learning and contextual understanding that transformer-based models offer.

7.2 OVERALL CONCLUSION

The experiments strongly suggest that while traditional models remain reliable and effective with structured data, **large language models (LLMs), especially when fine-tuned, unlock new potential** in predictive modeling.

1. Strength of Traditional Models:

- a. Models like XGBoost and ANN demonstrated strong performance, especially when paired with feature selection and preprocessing strategies. These models remain more computationally efficient and interpretable, which are desirable characteristics in production settings.

2. Superiority of Fine-tuned LLMs:

- a. The fine-tuned FLAN-T5 model significantly outperformed all others, reaching **94.49%** accuracy. This leap from **83–87% (LoRA-based)** to nearly **94.5%** validates the importance of domain-specific adaptation.
- b. Unlike benchmark models that rely on explicit feature engineering, LLMs have the ability to **learn latent patterns and semantic relationships** among questionnaire responses, especially when provided with full contextual data during training.

3. Viability for Real-World Applications:

- a. The strong performance of LLMs even with medium-sized datasets underscores their **practical applicability in clinical or diagnostic support systems**, particularly when questionnaires or surveys form the input modality.
- b. With increasing computational resources and open-source tools for fine-tuning, LLM-based solutions are becoming more accessible to healthcare data science teams.

7.3 FUTURE SCOPE

To further enhance diagnostic modelling using LLMs, the following extensions are proposed:

1. Multimodal Data Integration:

- a. Integrating other data sources such as imaging, electronic health records (EHRs), or genomics alongside questionnaires can significantly improve predictive power. LLMs can be adapted to handle **text-image or text-signal inputs** using cross-modal encoders.
 - b. This would help simulate real-world diagnostic reasoning more accurately, where doctors use visual, behavioural, and historical data together.
- 2. **Chain-of-Thought Reasoning:**
 - a. Implementing chain-of-thought prompting or multi-step reasoning allows the model to **explain its decision-making process**, improving transparency and interpretability.
 - b. This can be crucial in sensitive fields like healthcare where model interpretability is as important as accuracy.
- 3. **Prompt-Based Adaptive Diagnosis:**
 - a. Future iterations could use **dynamic prompting strategies** that personalize prompts based on patient profile, symptoms, or known history.
 - b. Adaptive diagnosis using few-shot or in-context learning can make models more responsive and context-aware without requiring full retraining for new symptoms or populations.

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